

OVERVIEW OF JENNINGS RANDOLPH LAKE OPERATIONS

North Branch Potomac
Sustainable Rivers Program Workshop

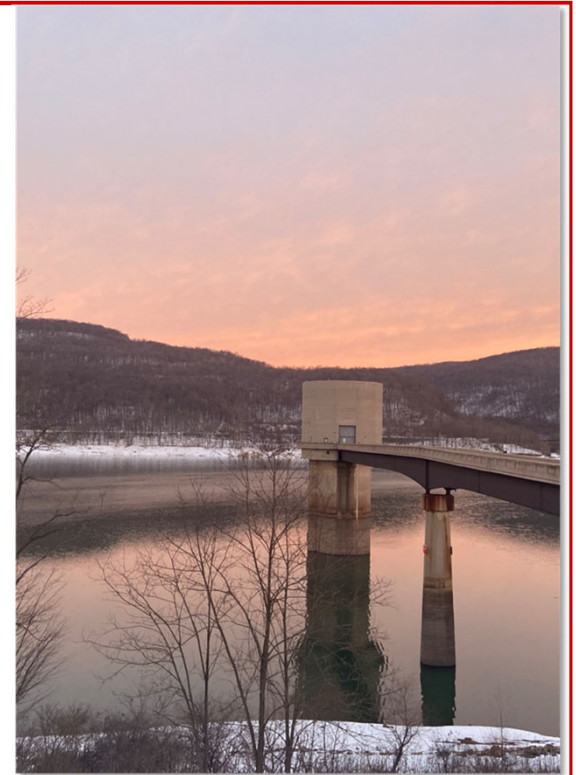
06 August 2025



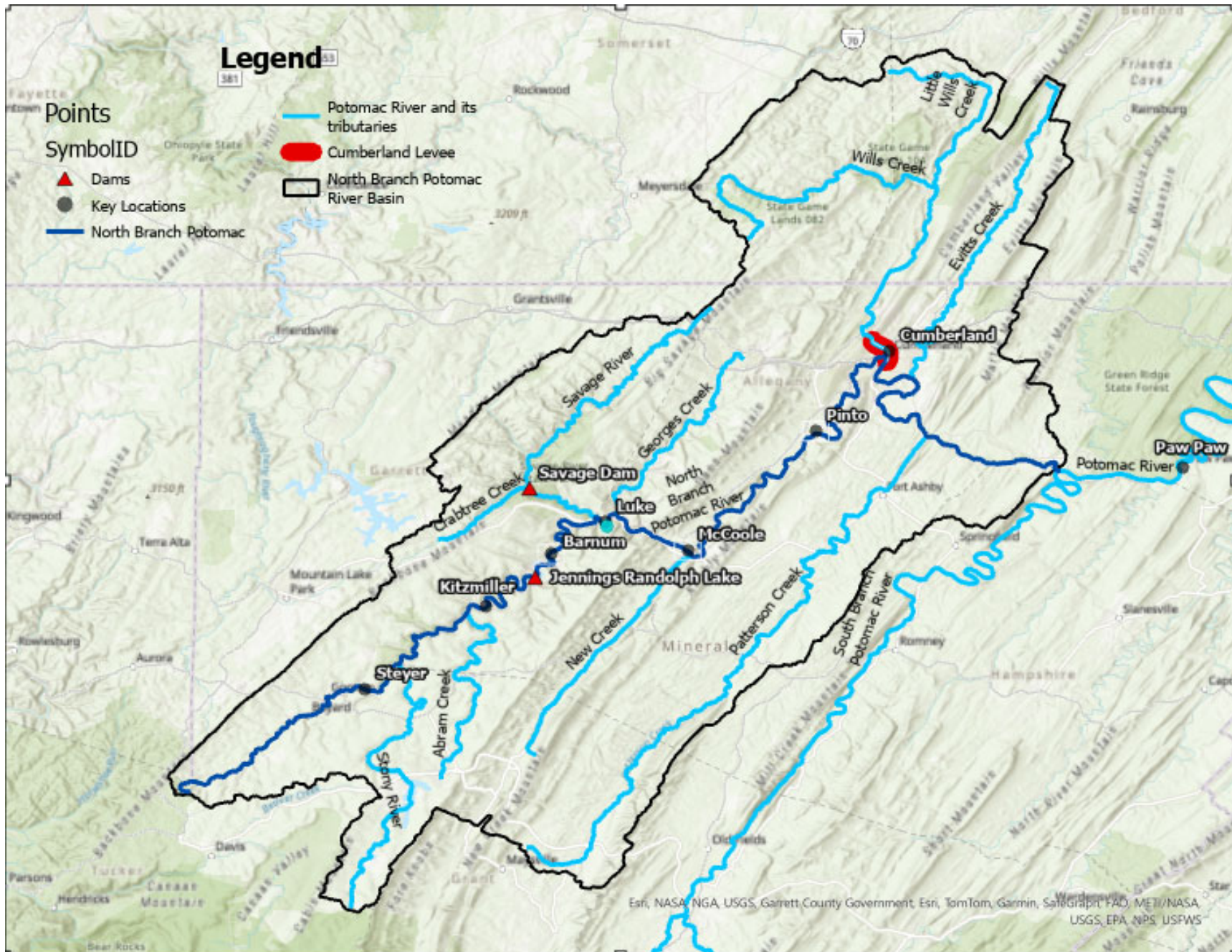
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AGENDA

- 01** NORTH BRANCH POTOMAC RIVER WATERSHED
- 02** NORTH BRANCH POTOMAC RESERVOIR PROJECTS
- 03** JENNINGS RANDOLPH REGULATION
- 04** WORKSHOP GOALS AND OBJECTIVES

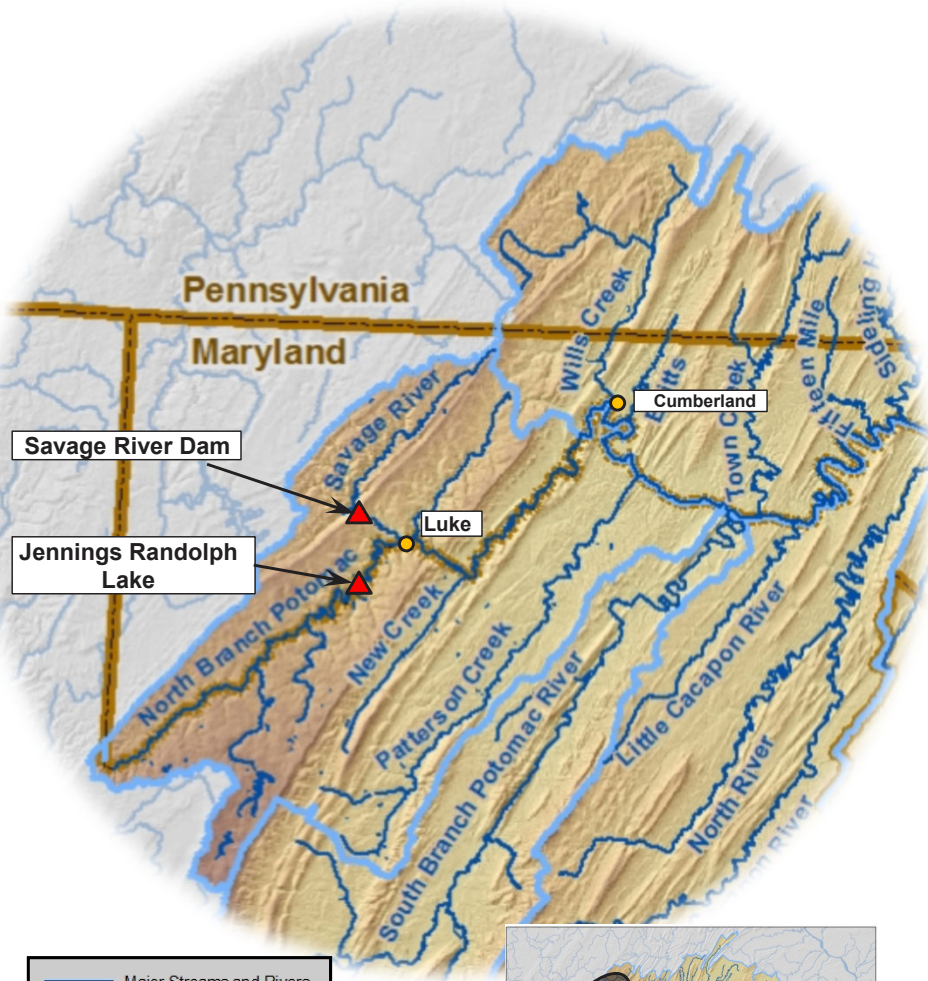


SUSTAINABLE RIVERS PROGRAM STUDY AREA



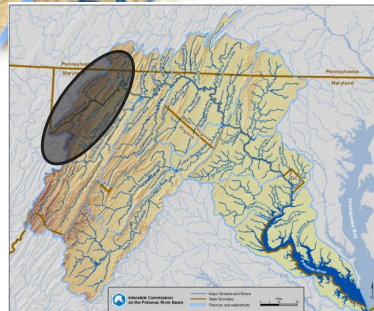
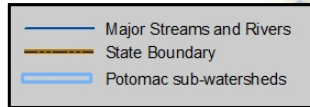
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NORTH BRANCH POTOMAC RIVER BASIN



Savage River Dam

Jennings Randolph Lake



Jennings Randolph Lake

- ❖ Location: North Branch Potomac River in Garrett County, MD & Mineral County, WV
- ❖ Designed & Constructed by the Corps of Engineers
- ❖ Construction completed in July 1981 (reservoir filled in 1982)
- ❖ Owned & Operated by the Corps of Engineers

Savage River Dam

- ❖ Location: Savage River in Garrett County, MD
- ❖ Designed by the Corps & partially constructed by the Works Progress Administration (WPA) (before WWII)
- ❖ Construction completed in 1952 by Corps of Engineers after WWII
- ❖ Owned, operated & maintained by Upper Potomac River Commission



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SAVAGE RIVER DAM



Physical Data

Drainage Area

105 sq. mi.

Maximum Dam Height

184 ft.

Storage

29,805 ac-ft (Top of Dam)

18,833 ac-ft (Conservation)

Outlet Configuration:

- Outlet Tunnel with Flood Gates
- Ungated Spillway



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JENNINGS RANDOLPH LAKE

Physical Data

Drainage Area
263 sq. mi.

Maximum Dam Height
296 ft.

Outlet Configuration

- Gated Spillway
- Selective Withdrawal Intake Tower with Quality Control Gates
- Outlet Tunnel with Flood Gates

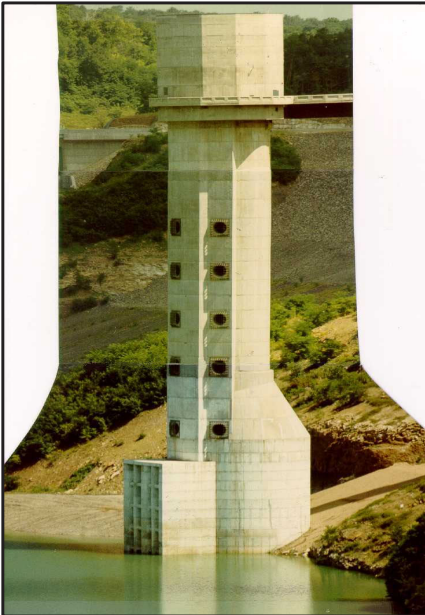


Storage Allocation

Flood Control Storage
34,448 ac-ft (28%)

Water Supply
40,194 ac-ft (33%)

Water Quality
50,009 ac-ft (39%)



OPERATIONAL OBJECTIVES

Savage River Dam



- ❖ Flood Control
- ❖ Low Flow Augmentation
- ❖ Water Quality Improvement
- ❖ Westernport Water Supply

Jennings Randolph Lake

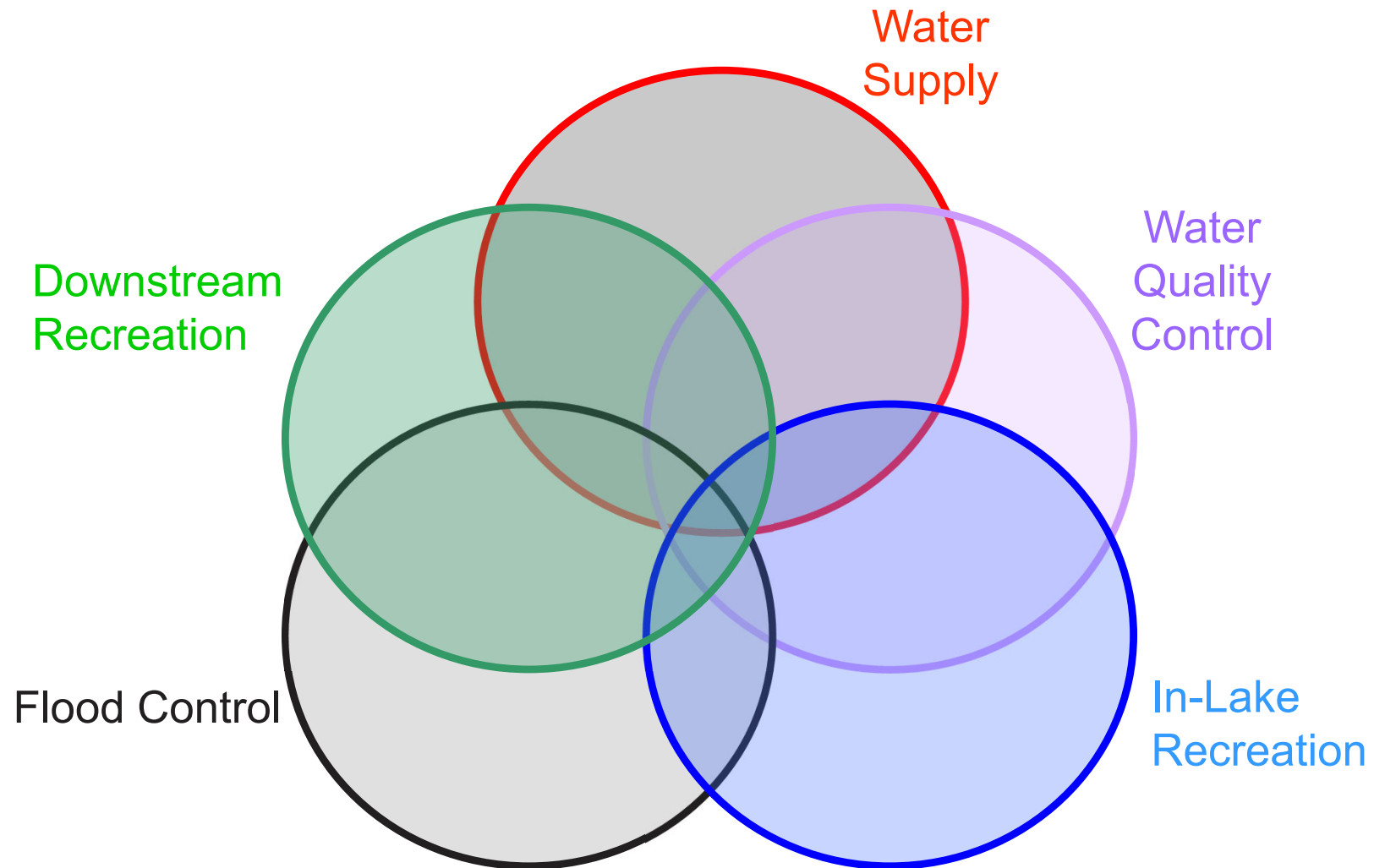


- ❖ Flood Control
- ❖ Water Supply
- ❖ Water Quality Control
- ❖ Recreation



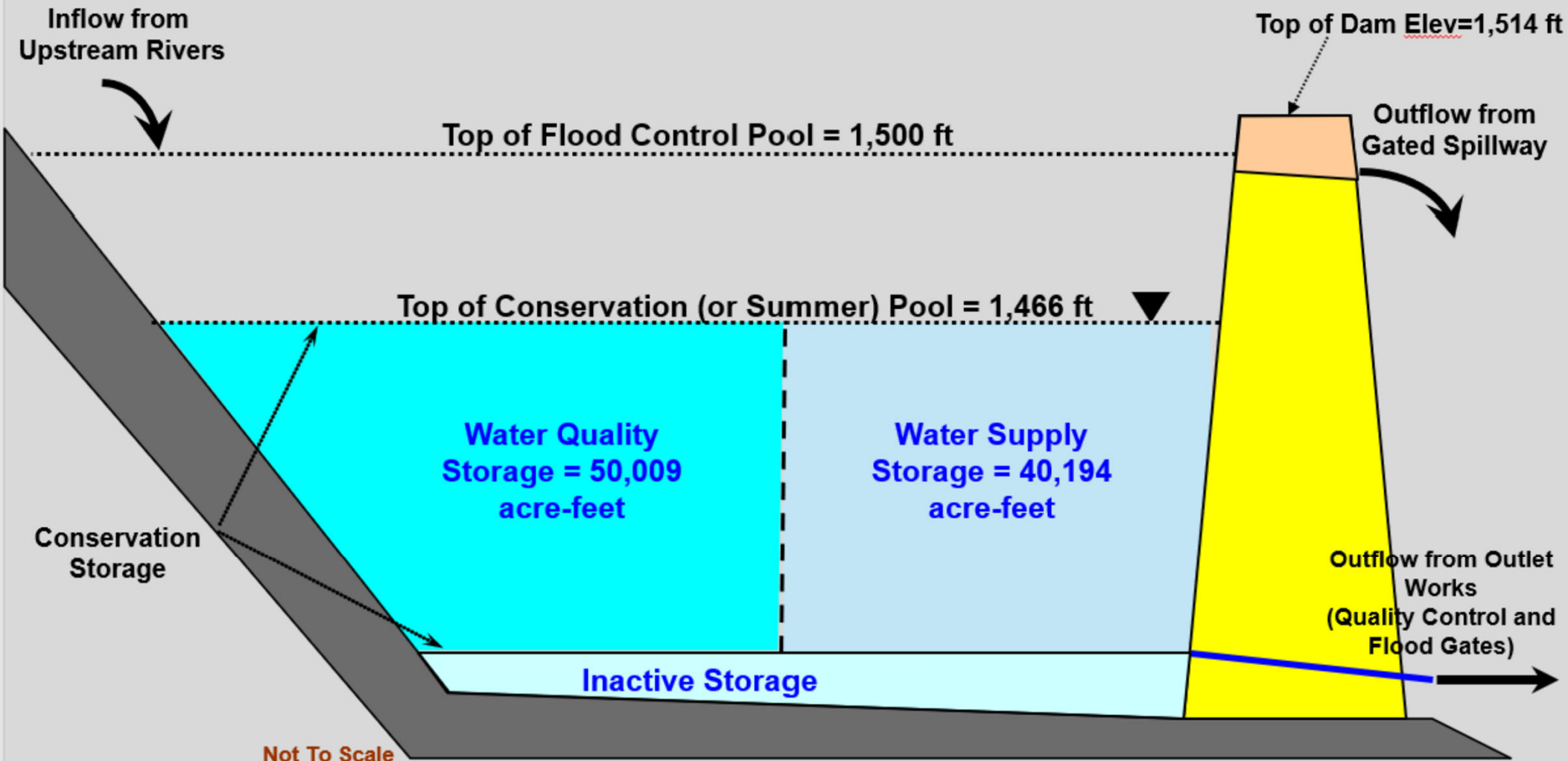
MULTIPLE PURPOSE PROJECTS

Complimentary, competing, & conflicting uses



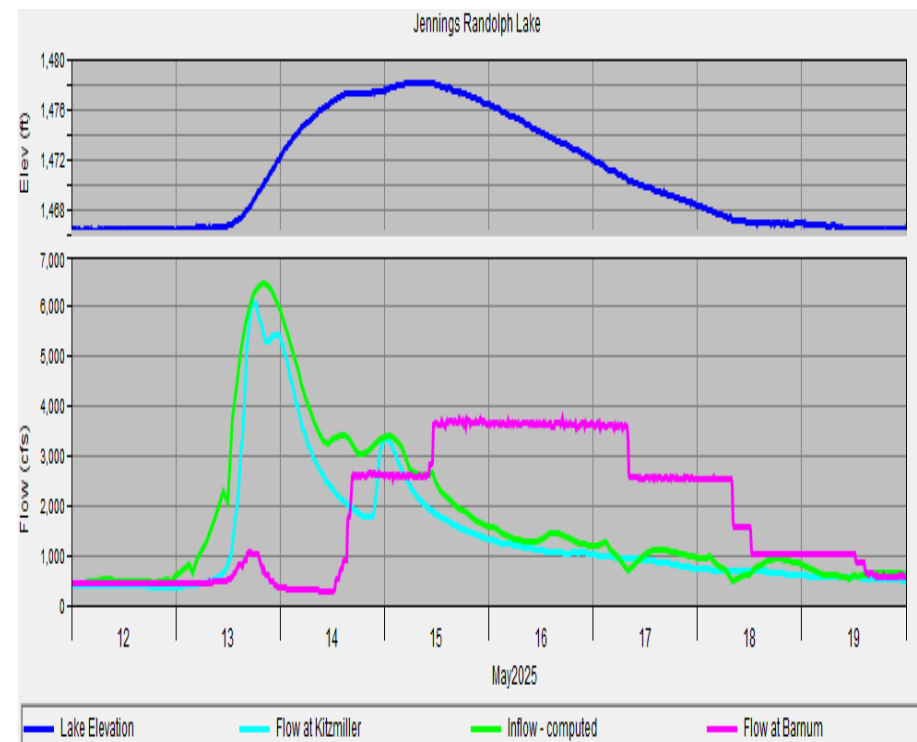
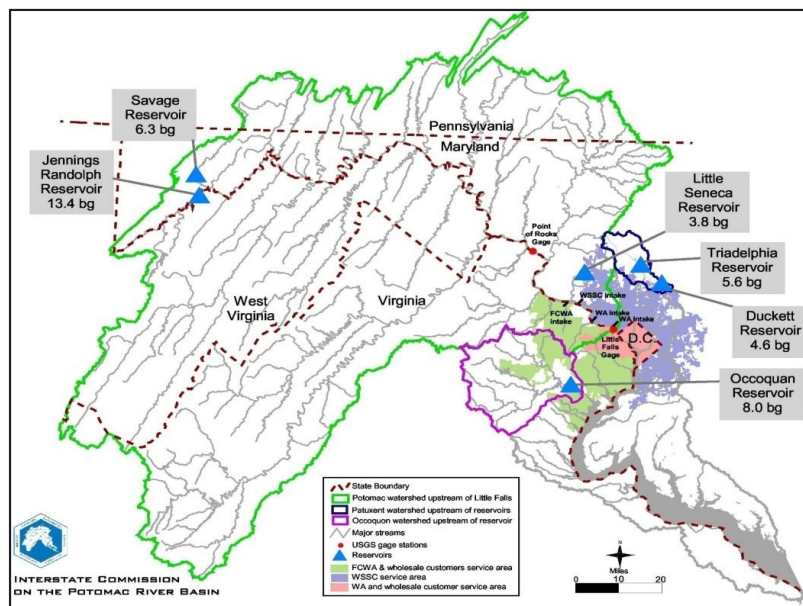
Jennings Randolph Lake

Lake Elevations & Storage Allocations



JENNINGS RANDOLPH LAKE – OPERATIONAL OBJECTIVES

- **Flood Risk Management**
 - Store water during flood events to reduce downstream river stages
 - Primary communities benefiting include Westernport, Piedmont, Keyser, Cumberland, Ridgeley and Oldtown
- **Water Supply**
 - Provide storage space (40,194 ac-ft) for Metro DC water supply
 - Make releases for water supply through coordination with ICPRB



JENNINGS RANDOLPH LAKE – OPERATIONAL OBJECTIVES

➤ Recreation

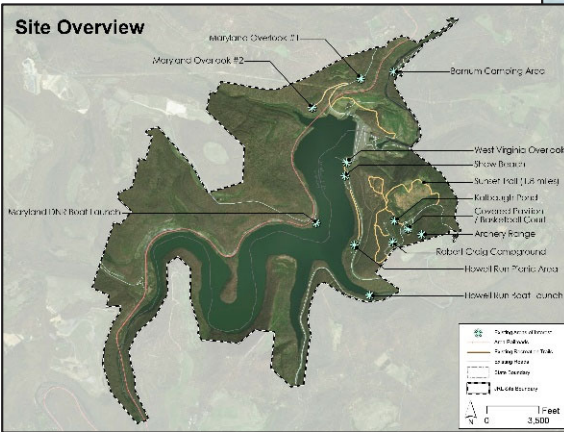
IN LAKE

- Boating
 - WV boat launch usable lake elevation >1445'
 - MD boat launch usable lake elevation >1420'
- Fishing
- Beach
 - Usable lake elevation >1455'

DOWNSTREAM

- Wade fishing
 - Preferred flow <300 cfs
- Float Fishing
 - Preferred flow 250-300 cfs
- Whitewater
 - 4 releases scheduled for every other weekend in Apr & May
 - Preferred flow 1000-1500 cfs

Site Overview



PROJECT LANDS

- Campground
- Picnic Areas
- Hunting
- Hiking

JENNINGS RANDOLPH – OPERATIONAL OBJECTIVES

➤ Water Quality

- Improve in-lake and downstream water quality
- 50,009 acre-feet (55.4%) of storage is allocated to water quality
- Historically used to manage releases for pH, water temperature, and overall pollutants.
 - Improve pH
 - Manage water temperature
 - Achieve uniformity in water quality releases
 - Dilute wastewater discharges from downstream municipal and industrial users
 - Augment North Branch flows during low flow seasons
 - Maintain a healthy downstream aquatic habitat



JENNINGS RANDOLPH – OPERATIONAL OBJECTIVES

➤ Water Quality

- Over time, change in strategies as watershed conditions have improved

NOW

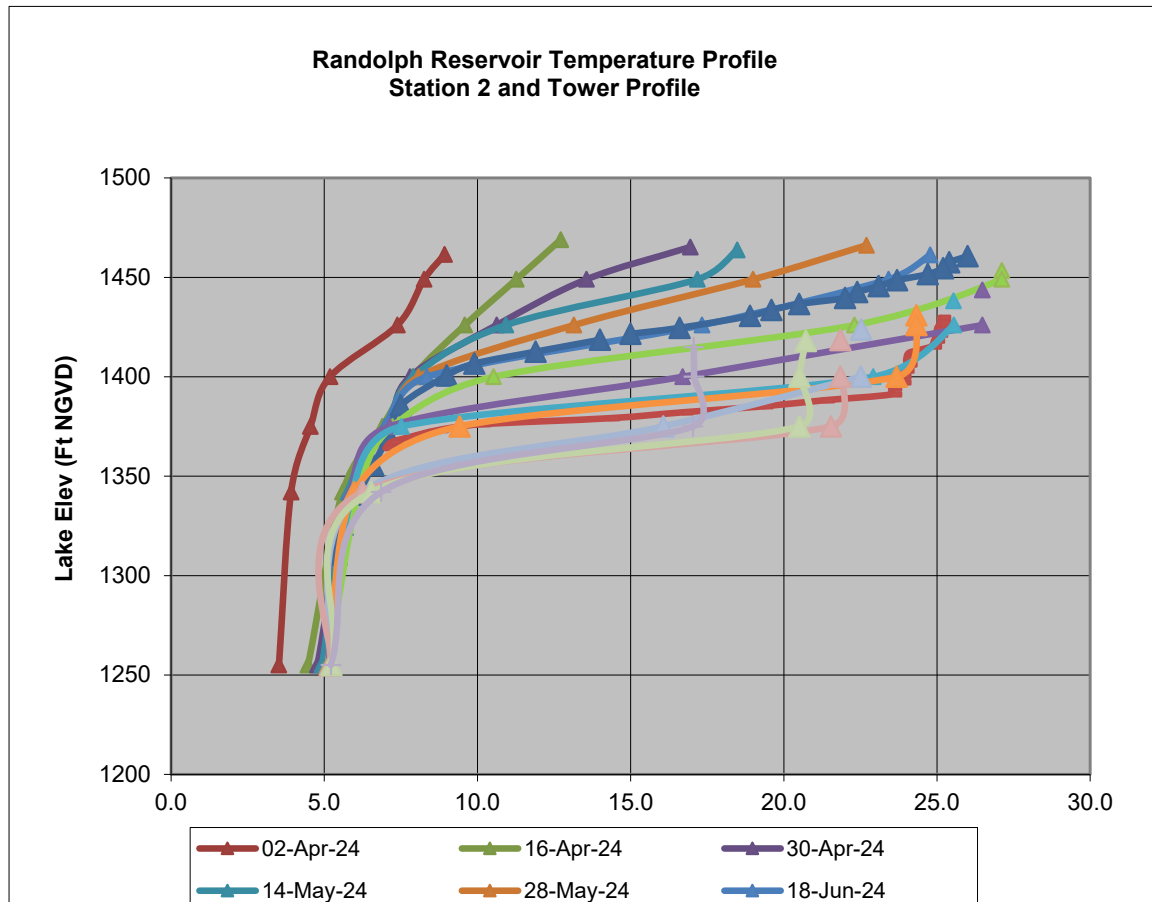
- Minimum flow release
 - Maintain a minimum of 120 cfs at Luke, combined with Savage release
 - Ideal flow at Luke of 300 cfs but
- Water temperature
 - Temperature of around 13°C is released from the project in the spring, summer, and fall
 - Conserve cold water to allow for cooler temperatures to extend into the fall
- Other Water Quality regulation factors:
 - Artificially Varied Flow Releases
 - Gas Supersaturation



JENNINGS RANDOLPH – WATER QUALITY REGULATION

➤ Selective Withdrawal System

- Five sets of ports located along the intake tower
- Allow for selecting and mixing water from various levels within the reservoir
- Provide a wide range of releases for downstream water quality



...view of the intake ports of the selective withdrawal system at Jennings Randolph Lake. The intake tower consists of two 7'2" x 16'4" flood gates located near the bottom / two 2' x 3' quality control (QC) gates, each with five ports. As we view the picture here, they are evenly numbered on the right, from top to bottom as P10,8,6,4,2. Selecting/mixing water from different levels in the lake provides a wide range of releasing quality water downstream of the dam.

JENNINGS RANDOLPH – ARTIFICIALLY VARIED FLOW (AVF) RELEASES

THE PROBLEM

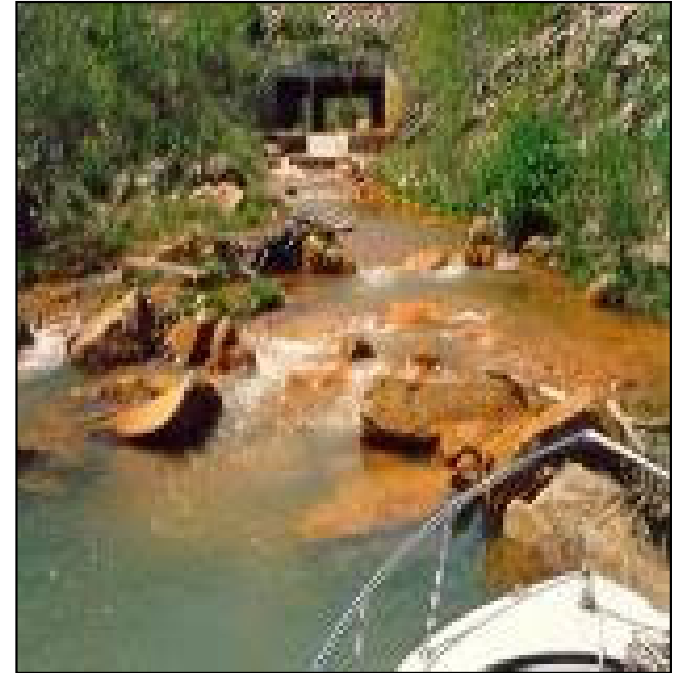
- Historic acid mine drainage & heavy industrial use in the North Branch Potomac River watershed
- Extremely low pH
- Suspended organic solids settling out onto the river bed
- Depletes dissolved oxygen, smothers benthic organisms

THE PLAN

- Starting in 1981 a program of reservoir induced AVFs for water quality enhancement was begun
- Deliberately surging stream flows from Jennings Randolph Lake to remove or reduce undesirable sediment accumulation from industrial pollution

RELEASES

- Held in late August & mid September
- Typical AVF event: 1,000 cfs for 24-36 hours
- Events are also used by whitewater recreators
- Re-evaluate benefits of these releases now that watershed WQ conditions have improved



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JENNINGS RANDOLPH – GAS SUPERSATURATION

THE PROBLEM

- In 1990, fish kills were observed in fish-rearing pens within the stilling basin at JRL when the project was discharging 4,200 cfs
- Determined that the fish mortality resulted from prolonged exposure to supersaturated gas

THE PLAN

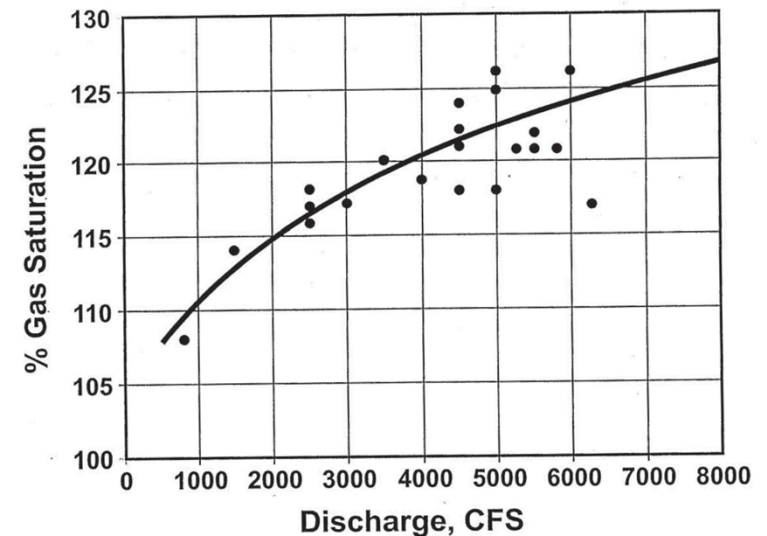
- Operational changes were made to reduce gas supersaturation by controlling the discharge rate and frequency of high-volume discharges.
- Fish pens were removed in 2007

RELEASES

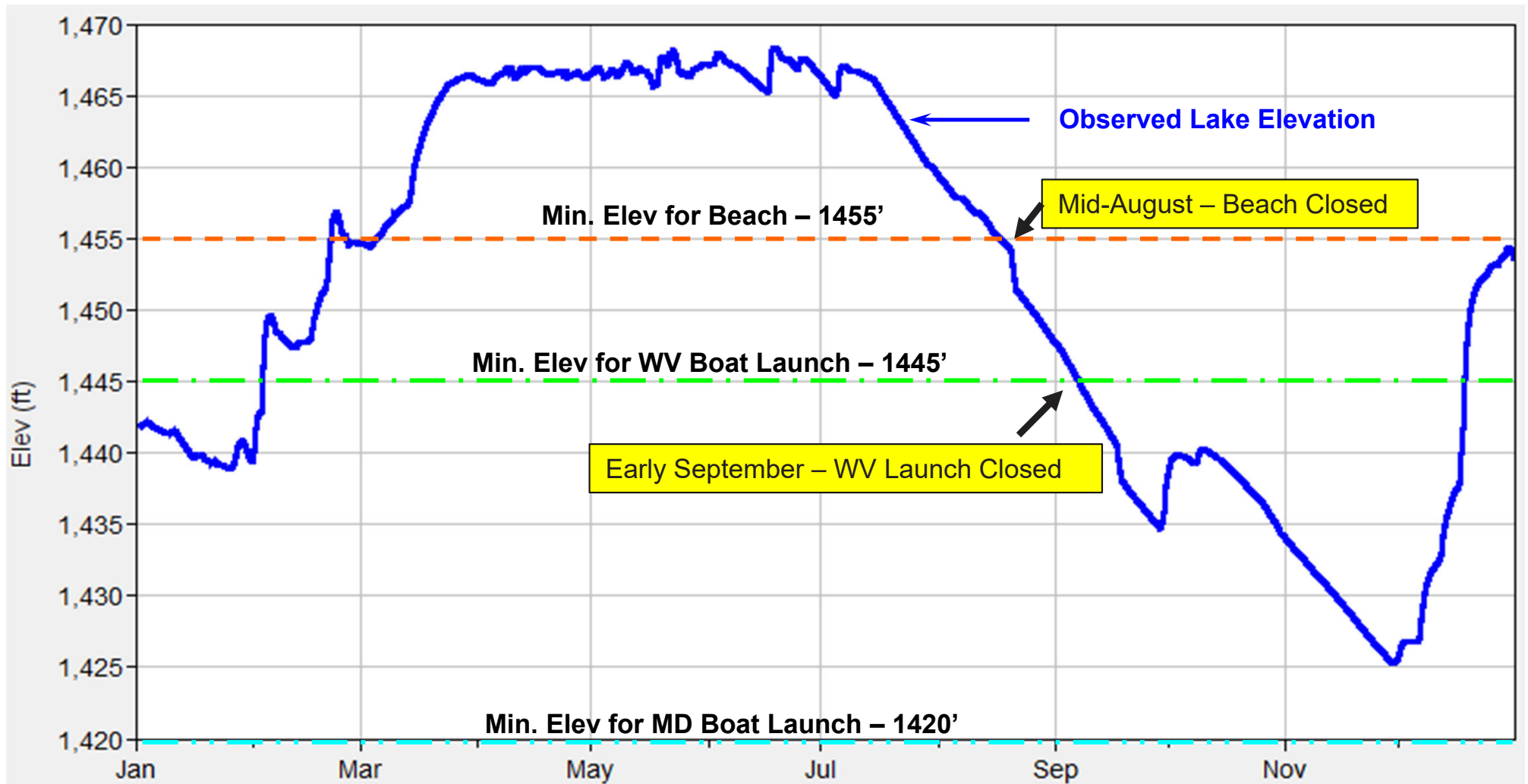
- Maintain discharges below 2,000 cfs
- 2,500 cfs is the upper limit
- If releases exceeding these amounts are needed, the duration is limited as much as possible



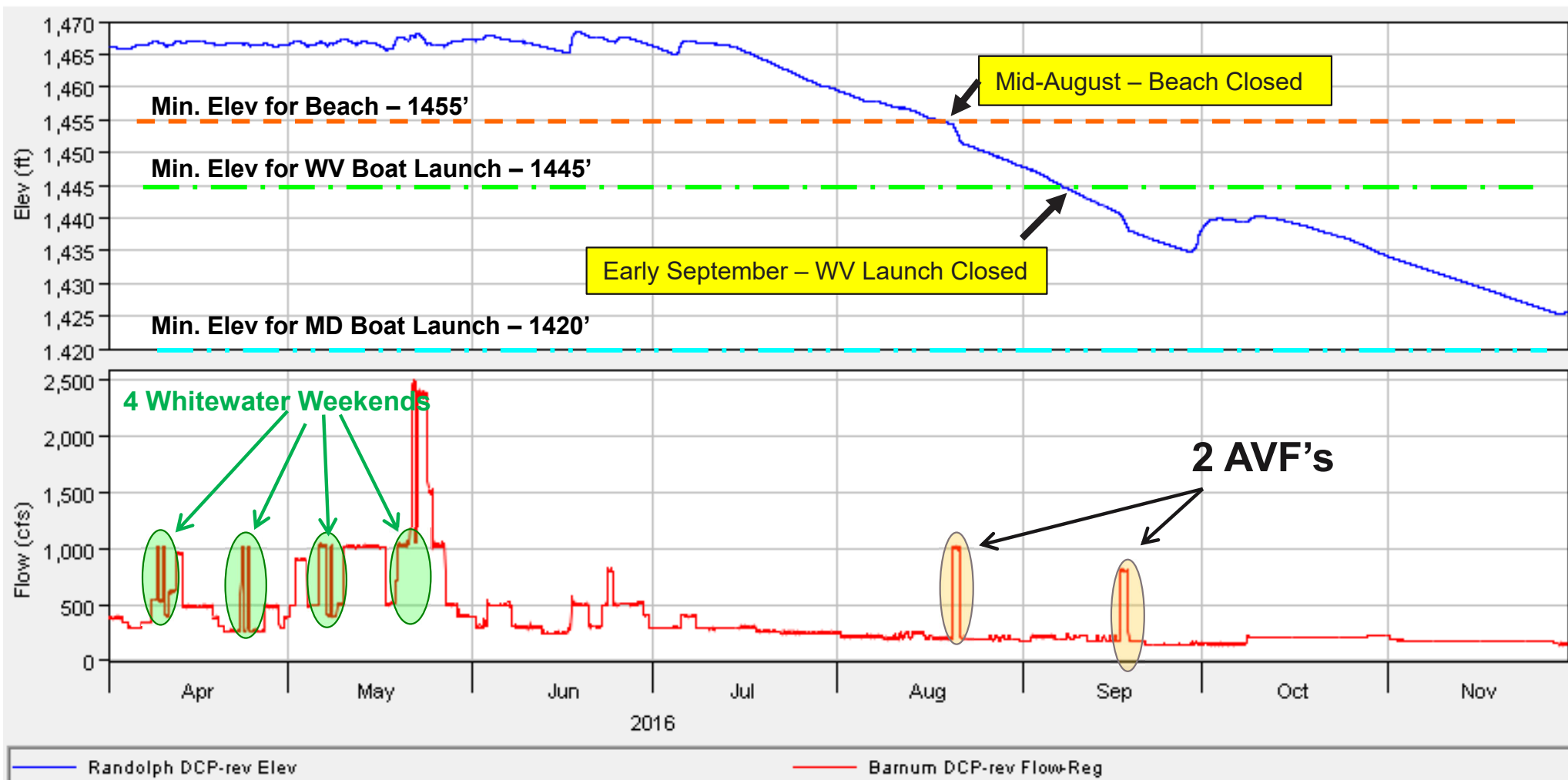
Jennings Randolph Lake



TYPICAL LAKE ELEVATION CURVE JENNINGS RANDOLPH LAKE



TYPICAL DOWNSTREAM ACTIVITIES JENNINGS RANDOLPH LAKE



NORTH BRANCH POTOMAC SRP & WORKSHOP OBJECTIVES

- Inform a re-evaluation of how water quality storage and releases from Jennings Randolph Lake are managed given the changed and improved watershed conditions post-construction
 - Re-evaluate current release strategies that are currently and/or historically utilized such as:
 - Artificially Varied Flow Releases
 - Maximized use of water quality storage (original release prescription)
 - Identify knowledge gaps/additional info needed



TRENDS IN REGULATION APPROACH

Project Purpose	Then...	Now...
Flood Risk Management	➤ Reduce outflow when downstream stages approach flood levels	➤ No change
Water Supply	➤ Release water for Metro DC area by request (through ICPRB)	➤ No change
Water Quality	➤ Manage to improve pH, conductance & DO ➤ Release as much water as possible for as long as possible ➤ Make occasional artificially varied flow releases	➤ Manage outflow for water temperature along with pH, conductance & DO ➤ Improve in-stream environmental habitat ➤ Make occasional artificially varied flow releases
Recreation (In-Lake)	➤ Camping, picnicking, boating	➤ Added fishing, swimming beach and additional boat launch
Recreation (Downstream)	➤ None	➤ Fishing (wade & float) ➤ Kayaking & canoeing (flat water and white water)

KNOWLEDGE GAPS

Flow effects in the North Branch Potomac	• Drought effects • Vegetation • Floodplain connectivity • Riparian forests • Ecological response to flow
Water quality	• Sedimentation in river • Improvements needed besides pH • DO concentrations in organic solid sediments on river bottom. • Changes since 2019 closure of Luke Paper Mill
Ecology	• Mussel/clam (bivalve) surveys • Effect of flows on invasive species
Groundwater	• Groundwater withdrawal effects • Groundwater dominant areas of the river, injecting cold-water



